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R-NEB: Accelerated Nudged Elastic Band Calculations by Use of Reflection Symmetry

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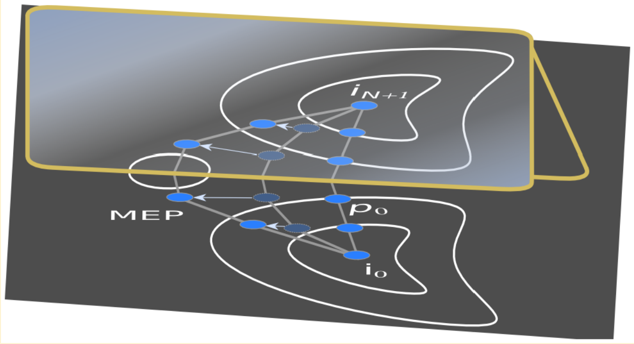
Supporting Information

ABSTRACT: Many activated processes in materials science, physics, and chemistry, e.g. diffusion processes, have initial and final states related by symmetry. Identification of minimum energy paths in such systems with methods such as nudged elastic band (NEB) can gain substantial speed up if the symmetry is exploited. The identification of minimum energy paths and transition states for such processes constitute a large fraction of the CPU-usage within computational materials science; much of which is in essence redundant due to symmetry. Paths with a reflection symmetry can be calculated using about half the computational resources, and the activation energy can, for some transitions, be estimated with high precision with a speed up factor equal to the number of images used in a standard NEB calculation. We present the formal properties required for a system to guarantee a reflection symmetric minimum energy path and an implementation to prepare and effectively speed up nudged elastic band calculations through symmetry considerations. Five examples are given to show the versatility and effectiveness of the method and to validate the implementation. The method is implemented in the open source package Atomic Simulation Environment (ASE) and contains internal methods to identify symmetry relations between the given end point configurations.



1. INTRODUCTION

For more than 20 years the nudged elastic band (NEB)[1−3] method has been one of the most used methods for finding minimum energy paths (MEPs) of thermally activated transitions. The NEB method has been refined in many ways since its original implementations. A significant improvement is the climbing image NEB that converges more closely on the maximum of the MEP, thus providing a more accurate estimate of the activation energy.[4] Also, algorithms for more efficient and faster NEB calculations have been developed to deal with a specialized set of systems[5−8] while others are more general.[9] Most recently, efforts have been invested in speeding up NEB calculations through machine-learning assisted algorithms.[10,11] But, to our knowledge, symmetry in the atomic configurations has not yet been exploited in NEB calculations in a systematic manner. In many transitions of interest, such as diffusion events in crystalline structures (vacancy, defect, and polaron migration), rotations of ligands on molecules, and even reactions at surfaces (surface migration), there is a symmetry relation between the initial and final configurations of the atoms. When the configurations are related by reflection symmetry, there is a corresponding symmetry in the potential energy surface (PES) with a mirror object placed between the two geometries. This guarantees that the energy barrier is symmetric around that object and it is then sufficient to map the MEP explicitly only on one side of the mirror object. The other half of the MEP can simply be found by applying the appropriate symmetry operations to the explicitly calculated part. Often, only the height of the energy barrier is of interest. In many such cases, where the path is sufficiently simple, a band with only one atomic configuration placed at the mirror object can be enough to converge to a precise estimate of the energy barrier. Rapid assessment of such barriers would be highly desirable for, e.g. Kinetic Monte Carlo simulations,[12,13] training of machine learning models, and cluster expansion based approaches.[14]

Here, we show how reflection symmetry can be used to greatly reduce computational effort in NEB calculations for such systems and present a user-friendly implementation of the method in the Atomic Simulation Environment (ASE).[15] The implementation could be further improved by making use of machine-learning approaches. The article is organized as follows: The NEB method is reviewed in section 2. In section 3 the formal requirements for reflection symmetry in the MEP are presented along with a discussion of the NEB algorithm and the generation of the initial path. Section 4 outlines the algorithm of the original NEB method and the suggested reflective NEB (R-NEB) and reflective-middle-image NEB (RMI-NEB) methods. In section 5 the details of the implementation are presented. In section 6 the results from

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five test problems are presented followed by the last section with some final comments.

2. NUDGED ELASTIC BAND METHOD

The NEB method is used to locate MEPs between two local minima on a PES. Given an initial guess of the path, an iterative optimization is carried out using some algorithm (e.g., BFGS,[16] CG,[16] velocity projection,[2,17] etc.). The initial path is often constructed by doing a straight line interpolation between end points, i.e. the two minima, to get a discrete representation of the path. However, a better guess is usually achieved by using the image dependent pair potential (IDPP) method.[18] In either case, the path is represented by a set of *images* where each image is characterized by the coordinates and spin directions of the atoms. The local minima are often related to the introduction of a *defect* into some otherwise perfect system, e.g. a vacancy, interstitial, displaced atom, polaron, adsorbate. Here, we will refer to the two local minima as the initial and final images i_0 and i_{N+1}, respectively. Likewise, the images with the defects at the initial and final position before structural optimization are referred to as the unrelaxed initial and final images i_0^u and i_{N+1}^u.

During the optimization of the path, only the N intermediate images are free to relax while the initial and final images are fixed. The key ingredient in the NEB method is the *nudging* of the forces. If each image were to follow the force acting on the atoms without modification, they would all slide down to the nearest local minimum providing no sampling of the interesting part. To avoid this only the force perpendicular to a local tangent of the path is included in the optimization. Additionally, spring-forces parallel to the local tangent are inserted between adjacent images. In this way the perpendicular force component ensures the images move toward the MEP while the spring forces control the distribution of the images along the path. Several methods for estimating the local tangent have been developed.[3,5,7]

3. FORMAL REQUIREMENTS OF REFLECTION SYMMETRY

Reflection Matrices. In a three-dimensional space, three types of reflections can be identified: Reflection with respect to a plane, line, or point. The latter two are the result of reflection with respect to two and three mirror planes, respectively. The matrices representing reflections are orthogonal and involutory, i.e., are their own inverse. They have a specific set of eigenvalues and a determinant of either 1 or −1. The eigenvalues and determinants used to identify valid reflection operations are summarized in Table 1.

Table 1. Characteristics of Reflection Matrices in Three-Dimensional Space

	Det(S)	eig(S)
plane	−1	{−1, 1, 1}
line	1	{−1, −1, 1}
point	−1	{−1, −1, −1}

Requirements of Reflection Symmetry. In the present implementation, the possibility that spin breaks any of the relevant symmetries is not taken into account. Working under this restriction, the symmetry of the PES is determined completely by the underlying symmetry of the atomic configurations. Consider an MEP connecting two symmetry equivalent atomic configurations, i_0 and i_{N+1}, related by a map S

$$i_0 = Si_{N+1} \tag{1}$$

where $S = TU$, in general, is a combination of rotation, U, and translation, T. For rotation, it is enough find a map between the equivalent geometries, $i_0 \sim Ui_{N+1}$, in the sense that all atoms have an equivalent atom in the other structure with an analogous local environment. However, in general, it will not result in configurations with the same absolute atomic positions.

Even though the end point atomic configurations are related by symmetry (point, line or plane), there is no guarantee that a path between the two is symmetric. To make sure that all the atomic configurations along the path have a mirror image, the atomic displacements must also be consistent with the symmetry operation. Take i_1 and i_N to be points on the path. Any such points can be reached from the end points by some atomic displacement according to a vector $\vec{R}$

$$i_1 = i_0 + \vec{R}_{0,1} \quad i_N = i_{N+1} + \vec{R}_{N+1,N} \tag{2}$$

If we choose $\vec{R}_{0,1}$ and $\vec{R}_{N+1,N}$ such that

$$\vec{R}_{0,1} = U\vec{R}_{N+1,N} \tag{3}$$

it follows that a path consisting of such points has reflection symmetry

$$Si_N = S(i_{N+1} + \vec{R}_{N+1,N}) = i_0 + \vec{R}_{0,1} = i_1 \tag{4}$$

Since any two symmetry equivalent atomic configurations must have the same energy, finding a path with reflection symmetry also means that the PES possesses the same symmetry and that any path—including the MEP—must respect this symmetry of the PES. In practice, we guess a path and search for a valid U which must fulfill eqs 1 and 3. After finding U, we know that the PES has this symmetry and that any valid path between i_0 and i_{N+1}, including the MEP, should have this symmetry. The result of the above is that *any path that respects the symmetry connecting two geometrically symmetry equivalent configurations always exhibits reflection symmetry.* It is also worth noting that any point belonging to the mirror object is guaranteed to correspond to a local extremum on any path that satisfies the reflection symmetry.

NEB Algorithm Conserves Symmetry. It is important to stress that the NEB algorithm preserves the reflection symmetry of a path. Given a path that consists of pairs of symmetry equivalent images, the true forces will be consistent with that symmetry. Since the tangent estimate and the spring force are based on vectors between adjacent images these vectors will necessarily also be consistent with the reflection symmetry. This in turn ensures that displacements along the nudged force conserves symmetry.

Linear Interpolation. A linear interpolation with equidistant images is characterized by N intermediate images that lie along the direction given by $\vec{R}_{0,N+1} = i_{N+1} - i_0$. This means that $\vec{R}_{0,1}$ and $\vec{R}_{N+1,N}$ as defined by (2) can be written as

$$\vec{R}_{0,1} = x(i_{N+1} - i_0) \quad \vec{R}_{N+1,N} = x(i_0 - i_{N+1}) \tag{5}$$

where $x = \frac{|\vec{R}_{0,N+1}|}{N+1}$. Thus, $\vec{R}_{0,1}$ is the invert of $\vec{R}_{N+1,N}$ and for eq 3 to be fulfilled, U must be a reflection performing that inversion

$$\vec{R}_{0,1} = U\vec{R}_{N+1,N} = -\vec{R}_{N+1,N} \tag{6}$$

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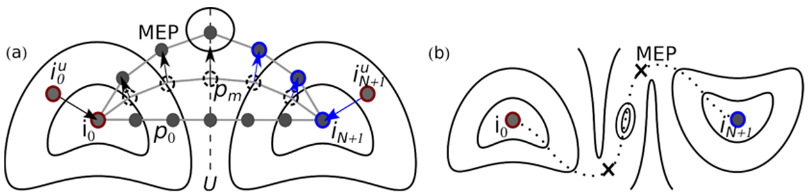

Figure 1. Illustration of the R-NEB method. (a) The solid black lines are contour lines of the potential energy surface. The circles represent images, i.e., atomic configurations. They are connected by gray lines to indicate a path. The bottom path, p_0, is the initial guess, the middle path, p_m, is an intermediate, m'th, step of the optimization, and the top path is the MEP. The dashed vertical line shows the mirror *plane* of the reflection U. Circles highlighted with a red border are images given as input while circles with blue borders are images constructed through symmetry operations. Black arrows represent calculated forces, and blue arrows represent forces obtained by symmetry. (b) Example of reflection with respect to a point. The inversion center is at the center of the figure, and the MEP is shown with the dotted line. The central contour lines mark a local minimum on the PES and the crosses indicate the maximum energy points on the MEP.

Thus, an equidistant linear interpolation results in a path respecting any reflection symmetry that is a map between the initial and final images and contains the mirror plane normal to the direction of the path. Furthermore, a linear interpolation is often an appropriate guess for these highly symmetrical systems. If modification of the linear interpolation is needed one has to make sure it is done in such a way that preserves reflection symmetry.

Spin Extension. Although we neglect cases where the spin configuration breaks the symmetry, the algorithm and implementation can be extended to handle spin even for systems with noncollinear effects. Such an extension of the present implementation is planned and could be integrated with the GNEB method.[8] The spin configuration, whether collinear or noncollinear, can simply be added as part of the state of every atom, which in the current implementation is only defined by its position in the cell. However, it is more complicated to make an educated guess for the spin arrangement along a given path so it may be necessary to introduce additional symmetry checks into the R-NEB algorithm.

4. ALGORITHM

In this section, we summarize the major steps involved in doing a traditional NEB calculation and present the procedure for the proposed R-NEB and RMI-NEB methods.

Traditional NEB. Besides doing the actual NEB calculation, a few preparatory steps are often done such that the set of calculations are

1. Define the initial image (e.g., introduce a defect, which is going to diffuse in a crystal, at its original position) and optimize the geometry.
2. Define final image and optimize again.
3. Interpolate between the initial and final geometry to get a path with as many points as is needed to get a good resolution of the energy barrier.
4. Optimize the path to converge on the MEP.

Reflective NEB. For symmetric systems we propose the following scheme for accelerated NEB calculations:

1. Introduce the defect at the initial position and optimize. This generates two configurations, the initial unrelaxed, i_0^u and the initial relaxed, i_0.
2. Introduce the defect at the final position (symmetry equivalent to the initial position) to get i_{N+1}^u and find the symmetry operation relating the initial unrelaxed geometry and the final unrelaxed geometry, $Si_0^u = i_{N+1}^u$.
3. Apply U to the changes made to the initial unrelaxed geometry during relaxation. Add this to the final unrelaxed geometry to get the final relaxed geometry, $U(i_0 - i_0^u) + i_{N+1}^u = i_{N+1}$.
4. Interpolate between i_0 and i_{N+1} to get N intermediate images.
5. Test that the path satisfies the reflection symmetry, see section 5 for more details.
6. Optimize the path with the usual NEB algorithm but carry out electronic structure calculations for only one-half of the path (including the middle image for odd N) and get energy and atomic forces of the other half by applying the appropriate symmetry operation.

To prepare an NEB calculation that includes symmetry, routines are needed for determining which symmetry operation relates a given pair of configurations and to test the symmetry of the initial guess of the path.

Reflective-Middle-Image NEB. In many cases, the path through which a transition occurs is not particularly important, only the activation energy, i.e., the height of the barrier, is of interest. The path is needed only to the extent that the highest maximum along the path can be identified. Since the final image is created to be perfectly symmetric with respect to the initial image, the resulting forces acting on the middle image of the path (for odd N) should be opposite and of the same magnitude (Figure 1). This means that the NEB algorithm will restrict the movement of the middle image to the mirror object during the optimization. This opens the possibility of doing NEB calculations where just a single geometry is optimized. However, it may well be that the maximum energy along the MEP is not at the mirror object. Also, one risks missing intermediate local minima along the MEP. Therefore, RMI-NEB calculation should only be carried out as a quick estimate of the barrier height or in cases where it is known that the maximum of the MEP is in the middle.

It is easy to check whether the middle image is at least a local maximum on the MEP. A dimer can be formed and rotated so as to find the direction of the MEP and the curvature obtained using finite differences.[19] An orthogonal dimer can then be used to ensure that the maximum corresponds to a first order saddle point on the energy surface. Then, the energy of this configuration can be used to estimate the activation energy of the transition, if it is known that the MEP does not have other, higher maxima. The RMI-NEB is potentially valuable for creation of databases where many similar structures are treated. In such studies a few MEPs could be calculated using the R-NEB method with a regular sampling and subsequently the activation

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energy could be evaluated for a larger number systems using the RMI-NEB, assuming the MEP has similar shape. Following the creation of activation energy databases, the results could, for example, be used for screening, machine learning, and multiscale modeling based studies. An RMI-NEB calculation follows the same preparatory steps for creating the final relaxed image (step 1−3) but at step 4 i_0 and i_{N+1} are interpolated by just a single image.

5. IMPLEMENTATION IN ASE

In this section, we present some details of the implementation in ASE. This includes a description of how configurations are compared, how translation and rotation symmetries are found, and how the symmetry of the path is checked. We also give some additional technical details and an overview of the features of the presented code.

Comparing Configurations. The matching of configurations is done atom by atom. Given two configurations to be matched, i_1 and i_2, consisting of atomic positions and chemical symbols, a search through i_2 is carried out for each atom in i_1. If the chemical symbols of the atoms match, the distance between the atoms $d = |r_2 - r_1|$ is calculated, where r_1 and r_2 are the positions of atoms belonging to i_1 and i_2, respectively. If d is less than a user controlled tolerance the atoms are considered to match. If a match is found for all atoms, the two configurations are considered to be the same.

Translations. Although not strictly needed, it is often useful and time saving to identify pure translations instead of combinations of rotation and translation when creating the relaxed final geometry. For pure translations

1. A list, $l_a = \{r_1^1, r_2^1, r_3^1, r_4^4, r_5^4, ...\}$, of all symmetry equivalent atoms, within the pristine simulation cell, is generated (using spglib[20]). r_x^y denotes the position of atom x which is equivalent to atom y.
2. l_a is used to generate a list of translation vectors, $l_{\vec{r}} = \{\vec{r}_{1,2}, \vec{r}_{1,3}, \vec{r}_{4,5}, ...\}$, where $\vec{r}_{1,2} = r_2 - r_1$. $l_{\vec{r}}$ contains all the translation vectors possible between pairs of equivalent atoms.
3. If $Ti_0^u = i_0^u + \vec{r} = i_{N+1}^u$, we have found a valid translation operation, T, between the initial and final configuration.

Note that by finding T, both a translation vector $\vec{r}$ and a list of atomic indexes is generated that can be used to match atoms when the configuration is translated.

Rotations. Rotations that are a map between two equivalent configurations are found as follows:

1. Identify the space group of the unit cell without defects and generate the appropriate list of symmetry operations, l_S, for the super cell.
2. Iterate through l_S:
   (a) If eq 1 is satisfied, that is $TUi_{N+1} = i_0$, then a valid U has been found.

As when finding pure translations, the comparison of the configurations results in a way to match atomic indices that can be used when applying U to get symmetry equivalent atomic displacements or forces acting on atoms.

Finding the Reflection of the Path. As discussed in the previous section we also need to test whether the path between the initial and final images is consistent with the reflection symmetry. We generate two lists of vectors; one with vectors moving stepwise forward and the other moving stepwise backward along the path. For odd N this is

$$p_f = \{\vec{R}_{0,1}, \vec{R}_{1,2}, ..., \vec{R}_{(N-1)/2,(N+1)/2}\} \tag{7}$$

$$p_b = \{\vec{R}_{N+1,N}, \vec{R}_{N,N-1}, ..., \vec{R}_{(N+3)/2,(N+1)/2}\} \tag{8}$$

If p_f and Up_b are equal elementwise, the path is symmetric with the given operation.

ASE Interface and Dependencies. The method above is implemented in Python. Although not the fastest programming language, it makes all the features of ASE available.[15] The code makes heavy use of the ASE features for manipulating atomic structures. Furthermore, ASE already contains an implementation of the NEB method which is just slightly modified to implement R-NEB and RMI-NEB. The code also depends on the python library spglib[20] for generating symmetry operators and lists of equivalent atoms within a single geometry. The implementation has the following features:

- Given initial unrelaxed, final unrelaxed, and initial relaxed configurations, the final relaxed configuration can be created.
- Given two configurations, the translation and rotation operators relating these can be found.
- Given a path defined by a set of atomic images, the reflection symmetries that apply to the path can be found.
- Given the appropriate symmetry operations, the NEB calculation can be carried out where only the symmetrically unique images are treated explicitly.

We stress that, given the input structures, the algorithm automatically identifies all the relevant reflection symmetries also for systems where these might be less intuitive to identify by visual inspection.

6. TESTS

Here we present five test cases where we benchmark the performance against the results of full NEB calculations. We focus on the speed up and the stability of the R-NEB and RMI-NEB methods. GPAW[21,22] was used to calculate the energy and atomic forces. Since we are interested in the performance of the method rather than the physics of these systems, further details about the DFT methodology is not given here but can be found in the Supporting Information. We used the FIRE[17] algorithm for optimization. All structures were relaxed until all forces were smaller than 0.02 eV/Å. For silicon and graphene, only the atomic degrees of freedom were allowed to relax while Li_2O_2 also was allowed to relax the lattice degrees of freedom (before introducing the defect). Seven intermediate images were included in the calculations of the silicon and graphene systems but five in the calculations of the Li_2O_2 system. A tolerance of 1×10^{-4} Å on the atomic positions was used. The results are summarized in Table 2.

The energy barriers calculated with the reflective and middle-image methods are less than 2 meV from the barriers calculated with the original NEB method. Thus, for all practical purposes the results are identical. Two barriers calculated with the RMI-NEB are off by more than a 2 meV. One is an example of an MEP where the transition state is not located in the mirror object and the other is an example of having end points which are not the global energy minimum of the MEP. This is described in more detail below.

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Table 2. Results for the Test Systems[a]

	precision [meV]			force calls			speed up	
	E_B^{std}	ΔE_B^{ref}	ΔE_B^{one}	N_{force}^{std}	N_{force}^{ref}	N_{force}^{one}	x_{force}^{ref}	x_{force}^{one}
Si	1057	0	0	343	196	43	1.8	8.0
C_6/V_C	804	0	61	434	248	44	1.8	9.9
C_6/across	356	0	2	371	216	38	1.7	9.8
C_6/along	400	0	31	357	196	39	1.8	9.2
Li_2O_2	1705	0	0	215	129	43	1.7	5.0

[a] E_B^{std} is the barrier calculated with the traditional NEB method. ΔE_B^{ref} and ΔE_B^{one} is the absolute error of the barrier estimate compared to the traditional NEB method for the R-NEB and RMI-NEB, respectively. Errors smaller than 1 meV are shown as zero error. x_{force}^{ref} and x_{force}^{one} are the ratios of the number of energy and force evaluations needed for convergence using the R-NEB and RMI-NEB methods, respectively, as compared to the traditional NEB method.

The number of force evaluations needed is expected to be roughly half for the reflective method

$$N_{force}^{std}\frac{N}{(N+1)/2} \quad \text{for odd } N \tag{9}$$

and for the middle-image method it is

$$N_{force}^{std}/N^{std} \tag{10}$$

where N^{std} is the number of intermediate images in the corresponding traditional NEB and N_{force}^{std} is total number of force evaluations needed using the traditional NEB method. However, especially for the RMI-NEB method, the number of force evaluations needed for convergence is even less than the expected N_{force}^{std}/N^{std} as the algorithm more effectively optimizes a single image than a set of N images.

For the presented systems the computational cost of doing the symmetry analysis is orders of magnitude lower than performing the corresponding DFT calculations. The CPU time spent on a single force evaluation is a factor 500−30 000 higher than what is spent on the symmetry analysis. Thus, when an electronic structure method, such as DFT, is used to evaluate the energy and atomic forces, the cost of the symmetry analysis is negligible and the reduction in computational effort is practically equal to the reduction in the number of energy and force evaluations; see eqs 9 and 10. Below, we describe the structure, symmetry, and transition path of the systems studied.

Silicon. The migration barrier for a silicon vacancy in silicon crystal was calculated. The crystal has diamond structure with a unit cell belonging to space group Fd$\bar{3}$m (227). The primitive unit cell has two atoms which are equivalent by symmetry. The vacancy is introduced into a 3 × 3 × 3 supercell with 54 atoms (Figure 2). The initial and final positions of the vacancy are neighboring sites. The vacancy moves along the [111] direction.

The symmetry analysis finds six maps between the initial and final images of which four are reflections. A path created from linear interpolation preserves all four reflections

$$U_1 = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \quad U_2 = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

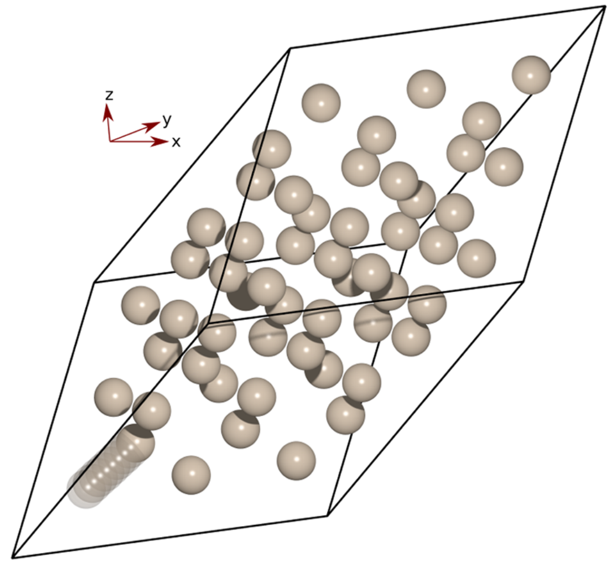

Figure 2. Silicon with a vacancy. The path is illustrated by the transparent atoms.

$$U_3 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}, \quad U_4 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

U_1, U_2, and U_3 represent reflections with respect two mirror planes, i.e. reflection in a line. U_4 is a point reflection in the point common to all of the mirror planes represented by U_1, U_2, and U_3. U_1 is reflection with respect to the $(1\bar{1}1)$- and (010)-plane, U_2 is reflection with respect to the $(11\bar{1})$- and (001)-plane, and U_3 is reflection with respect to the $(\bar{1}11)$- and (100)-plane. The energy barrier is bell shaped with the maximum, a first order saddle point, at the central image.

Graphene. The second test system is graphene and calculations are carried out for vacancy migration and two different diffusion jumps of an intercalated lithium atom (Figure 3). The primitive unit cell of graphene is hexagonal with two identical atoms and belongs to space group *P*6/*mmm* (191).

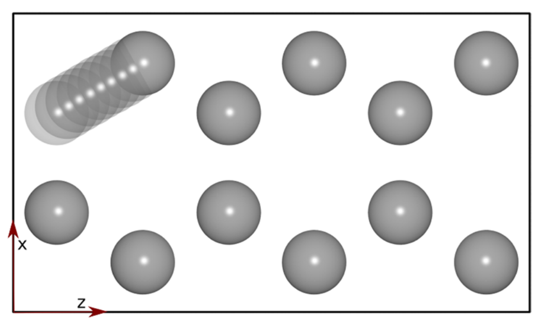

Figure 3. Graphene with a vacancy. The path is marked by the transparent atoms.

Carbon Vacancy. In this case an out-of-plane lattice constant of $b = 7$Å is used and a 2 × 1 × 3 supercell, i.e., a 2 × 3 repetition in the plane. The vacancy moves to a neighboring C-site. We find four valid maps between the initial and final images which are all reflections.

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The analysis of the path excludes two operations so that we are left with two valid reflections

$$U_1 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad U_2 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

U_1 represents reflection with respect to the (100)- and (001)-plane. In U_2 the trivial mirror plane of the two-dimensional graphene sheet is added to the mirror planes of U_1. The standard and reflective NEB calculations identify two energy minima on the MEP 61 meV lower in energy than the initial and final images; see Figure 4. The result is that the RMI-NEB misses the global minimum of the MEP and the deduced energy barrier then has an error equal to the difference between the energy of the end points and the global minimum.

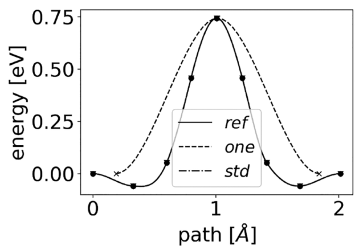

Figure 4. Calculated barriers for a carbon vacancy jump in graphene.

Intercalated Lithium Moving Across a C−C Bond. Here, an out-of-plane lattice constant of $b = 3.7$Å is used and a $2 \times 2 \times 3$ supercell. The Li atom has its equilibrium position at the center of a hexagonal C-ring. Neighboring sites are chosen for the initial and final states such that the Li travels across a single C−C bond; see Figure 5. We find three valid maps (not counting the identity) between the initial and final images that are all reflections. The analysis of the path excludes one operation so that we are left with two valid reflections

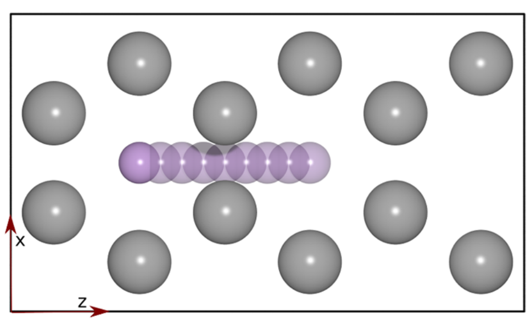

Figure 5. Intercalated lithium in graphite moving across a C−C bond. C atoms are gray and lithium is purple. The path is marked by the transparent atoms.

$$U_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad U_2 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

U_1 is reflection with respect to the (001)-plane, and for U_2 the (100)-plane is added on top. Note that all of the path belongs to the (100) mirror plane effectively confining the images to this plane during optimization of the path. This is another example of a bell shaped energy barrier as can be seen in Figure 7 together with the other energy barriers for lithium atom diffusion in graphite.

Intercalated Lithium Moving along a C−C Bond. An out-of-plane lattice constant of $b = 3.7$Å is used in this case and a $4 \times 2 \times 2$ supercell. Next nearest neighboring sites are chosen for the initial and final states such that the Li travels along a C−C bond when following the shortest path (Figure 6). We find three valid maps between the initial and final images that are all reflections. The analysis of the path excludes one operation so that we are left with two valid reflections

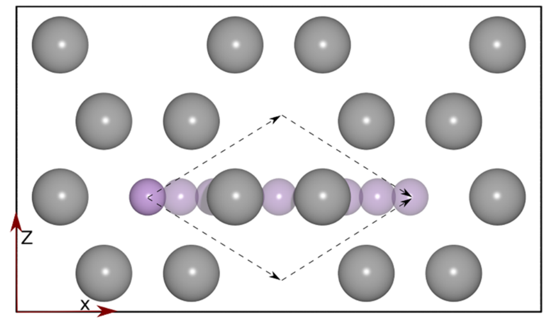

Figure 6. Intercalated lithium atom in graphite moving along a C−C bond. C atoms are gray and the Li atom purple. The path is marked by the transparent atoms. The true MEP is shown by the dashed lines with arrows.

$$U_1 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad U_2 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

U_1 is reflection with respect to the (100)-plane, i.e. the mirror plane reflecting the path. U_2 additionally includes the (001)-plane to which all of the path belongs. This an example of a path where the maximum along the MEP is not placed on the mirror object; see Figure 7. Since the middle image is fixed (by symmetry) to move only in the mirror planes it will never find the first order saddle point on the energy surface. A dimer calculation[19] would quickly reveal that the central image is trapped in a local minimum. Another thing to note is that the true MEP for these end points goes through one of the C-rings above or below the found path. Jumps across C−C bonds are preferred over a jump along a C−C bond since the energy barrier is lower. However, the images of the chosen initial path all lie in the (001)-plane. The symmetry around this mirror plane makes

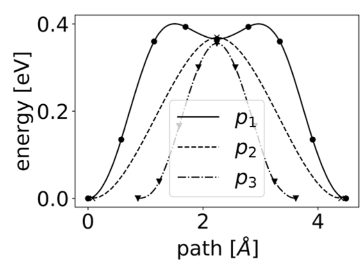

Figure 7. Calculated energy barriers for lithium atom moving in graphite. p_1 and p_3 are calculated with the R-NEB method for the Li atom moving along and across a C−C bond, respectively. p_2 is calculated with the RMI-NEB for the Li atom moving along a C−C bond.

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sure that forces taking the images toward the true MEP effectively cancel out. This situation can be avoided by checking whether the initial path lies completely in any of the mirror objects defined by the symmetry operations linking the initial and final images. If this is the case the initial path can easily be modified slightly in order to break the symmetry. If the mirror object coincides with a local minimum the images will find their way back and no harm is done but if, as in this case, part of the mirror object lies on a local maximum the images will move further away.

Li_2O_2. The last test system is Li_2O_2, which is a discharge product at the cathode of $Li-O_2$ batteries.[23] Since Li_2O_2 is a wide band gap material, which severely limits the discharge depth of the battery, it is important to determine whether sufficient charge-transport can be achieved through alternative channels, e.g. through migration of charged defects. One such charged defect, the negative lithium vacancy, is chosen in this example.

Li_2O_2 has a unit cell with eight atoms: four equiv O atoms and two different types of Li atoms. The space group is $P6_3/mmc$ (194). We work with a 2 × 2 × 1 supercell where a negative Li vacancy is introduced (a neutral vacancy would additionally introduce polarons into the system). The vacancy moves between two layers in the structure; see Figure 8. We find four

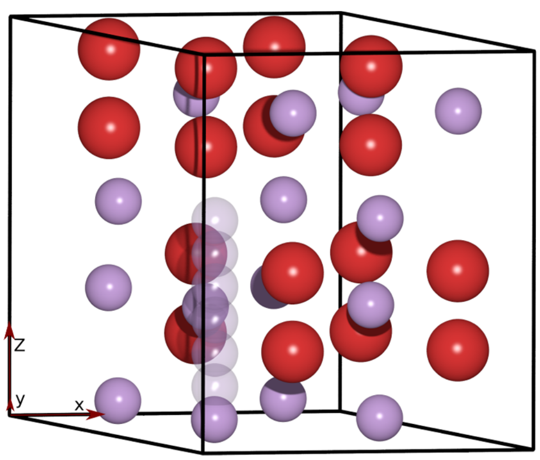

Figure 8. Li_2O_2 with a negatively charged lithium vacancy. O atoms are red, and Li atoms purple. The path is marked by the transparent atoms.

symmetries connecting the initial and final images that are all reflections. The movement associated with the path eliminates all but one symmetry

$$U_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

U_1 is reflection with respect to the (001)-plane. The energy barrier is bell shaped.

7. CONCLUSION

We have shown how reflection symmetry of various sorts can be used to speed up an NEB calculation of a minimum energy path using the R-NEB method, and in some cases obtain an estimate of the activation energy of a transition without even finding the whole path by using the RMI-NEB method. Any path, including the MEP, which is consistent with the symmetry connecting two symmetry equivalent end point configurations always exhibits a reflection symmetry. We demonstrate that the NEB algorithm preserves reflection symmetry so if the calculation is started with a path that is consistent with the symmetry, such as a linear interpolation, the optimization can be accelerated by carrying out the calculations for only one-half of the path.

The implementation we describe is a user-friendly extension to the ASE code that is able to find the symmetry relations between atomic structures and use them to perform NEB calculations at roughly half the computational cost or less.

The symmetry analysis part of the implementation can be used to tell whether two atomic structures are equivalent and which symmetry operations are valid maps between the two. Once one of the symmetry equivalent structures is relaxed this can be used to create the other one by applying the symmetry operation to the structural changes that occur in the relaxation.

A band of images can be analyzed to test whether they define a path consistent with any reflection symmetry connecting the initial and final images. The resulting symmetry and set of images representing an initial path can be passed to an NEB algorithm which will only perform force evaluations on the images on one side of the mirror object, thus saving roughly half the computational cost sacrificing practically no precision compared to the original NEB method. We show that, when symmetry is present, one can often get away with just a single image placed midway between the two end images and still get a good estimate of the energy barrier. This approach should, however, be used with care and a dimer calculation carried out to ensure that a first order saddle point has been identified. When it is applicable, it gives a very significant speed up and we expect that it will turn out to be useful when databases for activation energies for similar transitions in similar systems are generated.

The symmetry analysis is 500−30 000 times faster than a single electronic structure calculation, such as DFT. This difference in computational cost ensures that the speed up (practically) depend exclusively on the number of images we can avoid treating explicitly in the NEB calculation. Additionally, several symmetry checks could be performed during the R-NEB and RMI-NEB calculations at negligible computational cost. However, the tests we have made indicate that this is not needed as long as reasonable calculation parameters are used for getting the forces. Furthermore, if the assumed symmetry is broken, it is usually obvious by inspection of the path.

We continue work to include symmetry treatment of spin polarized systems and believe that the widespread use of the method could significantly reduce the amount of computational resources spent on finding MEPs. In order to get optimal performance, the method should be integrated or interfaced with other methods for speeding up NEB calculations.[9−11]

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jctc.8b01229.

Computational details, calculated energy barriers, and calculation timings (PDF)

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Notes

The authors declare no competing financial interest.

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